Week 2 – Data Pipeline Orchestration

London Success Academy Joint Internship Programme
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Week 2 Scenario

Company: MarketFlow Property Intelligence

UrbanNest has now grown and merged with another analytics company called MarketFlow Property Intelligence.

The company now receives daily property data updates from multiple real estate agencies.

Manually running data pipelines is no longer efficient. The company wants to automate the workflow using orchestration.

You have been hired as a junior data engineer to help automate the data pipeline.

Your responsibility is to design a Directed Acyclic Graph (DAG) to control the workflow.

Week 2 Learning Objectives

- Workflow orchestration
- DAG design
- Pipeline dependencies
- Data validation
- Monitoring concepts

Workflow Scenario

Daily property data arrives in S3. The system must automatically perform the following steps:

- Validate data
- Run ETL transformation
- Store processed datasets
- Update analytics tables

Week 2 Tasks

Task 1 – Understand DAG Structure

Example pipeline:

Data Upload

Data Validation

Transformation

Data Storage

Analytics Query

Task 2 – Design DAG Workflow

Create DAG nodes:

- task_1_validate_data
- task_2_clean_data
- task_3_transform_dataset
- task_4_store_parquet
- task_5_trigger_analytics

Task 3 – Build Pipeline Logic

Validation checks:

- Null values
- Duplicate records
- Data format validation

Task 4 – Create Monitoring Strategy

Design monitoring alerts for:

- Pipeline failures
- Missing data
- Job execution errors

Week 2 Deliverables

- DAG architecture diagram
- Pipeline workflow documentation
- Validation logic explanation
- Example monitoring plan

Skills Developed

Technical

- Pipeline orchestration
- DAG architecture
- Workflow automation
- Data validation strategies

Non-technical

- Systems thinking
- Dependency mapping
- Error handling strategies